



MEDALLION

Architecture

Designing a Medallion Architecture from Raw PC Sales
Data to Business Intelligence



TSHIKOSI
TSHIFHIWA



INTRODUCTION

- Medallion Architecture is a modern data design pattern used to organize data into progressive layers that improve data quality, reliability, and business value as data moves through each stage.
- Popularized by Databricks for Lakehouse and cloud-based data platforms
- Structures data into Bronze, Silver, and Gold layers
- Solves challenges like inconsistent transformations, poor quality, and duplicated logic
- Creates a clear path from raw data to business-ready insights





CORE LAYERS

Bronze Layer (Raw Data)

- Stores raw data exactly as received from source systems
- Minimal or no transformation
- Preserves historical records for traceability
- Supports reprocessing if downstream issues occur
- Example: raw PC sales csv

Silver Layer (Cleaned Data)

- Cleanses and standardizes Bronze data
- Removes duplicates and fixes nulls
- Applies validation and business rules
- Integrates datasets for consistency
- Example: validated customer profiles and standardized product data

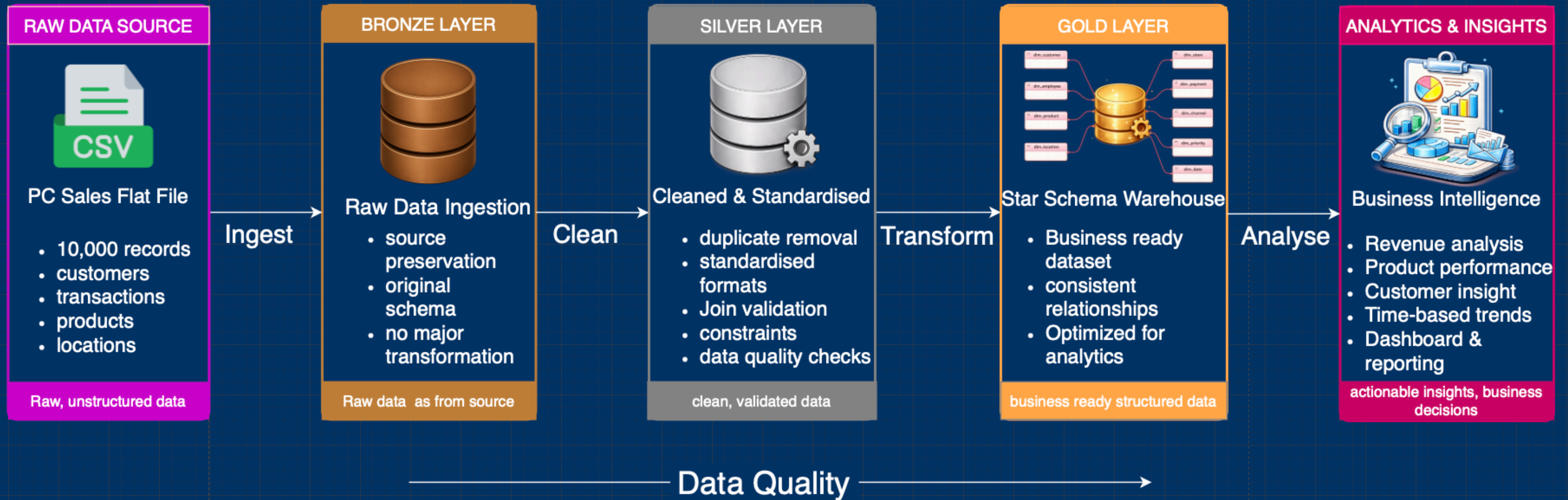
Gold Layer (Business-Ready Data)

- Curated for analytics, dashboards, and decision-making
- Aggregated and optimized for performance
- Supports star schemas, KPIs, and reporting
- Example: revenue dashboards, sales trends, and customer insights





PC Sales Medallion Architecture





EVALUATION

Benefits

- Improves data quality at each stage
- Enhances scalability for large datasets
- Supports data governance and compliance
- Preserves source data for auditability
- Enables faster reporting and dashboard performance
- Reduces complexity through structured separation
- Increases reusability of cleaned datasets

Advantages

- Clear data lineage from source to insights
- Better troubleshooting and error recovery
- Flexible for multiple analytics use cases
- Optimized Gold layer for BI tools
- Strong governance and compliance support
- Works well with Snowflake, BigQuery, Databricks, and Azure

Disadvantages

- Higher storage costs due to multiple layers
- Requires more engineering and maintenance
- Can be overly complex for small projects
- Slower for quick prototypes due to staged processing





Medallion vs Traditional Architecture

Traditional ETL

- Raw data often transformed directly into warehouse
- Limited raw data preservation
- Lower governance and flexibility
- Faster for simple projects but less scalable

Medallion Architecture

- Multi-layer refinement (Bronze → Silver → Gold)
- Full raw data preservation
- Strong governance and quality control
- Better scalability and long-term reliability

Key Difference

- Traditional = Direct transformation for immediate use
- Medallion = Progressive transformation for trusted, scalable insights





How Medallion Architecture Reduce Costs

Platforms like Microsoft Fabric and Databricks showcase how Medallion Architecture enhances modern data transformation and management:

- Microsoft Fabric unifies data warehousing, engineering, analytics, machine learning, and governance in one ecosystem for simplified management and predictable costs
- Databricks leverages Delta Lake and distributed processing to support scalable modern data lakes and AI-driven solutions

Both platforms reduce fragmented infrastructure costs by consolidating key data functions into a single, efficient ecosystem





CONCLUSION

- Medallion Architecture is a scalable, structured, and modern data engineering framework.
- Ideal for projects like PC sales analytics to transforms raw PC sales data (customers, products, payments, and transactions) into structured, business-ready intelligence
- Supports accurate sales analysis, customer insights, product performance tracking, and regional reporting
- Enables star schema modeling, executive dashboards, and scalable enterprise BI solutions
- Preserves historical data while improving quality, governance, and long-term strategic decision-making
- Provides a future-ready foundation for modern data platforms





Thank You



Tshikosi Tshifhiwa



tshikositshifhiwa4@gmail.com



+27 (60) 741 0085